

Economics Formula Sheet

Every equation and decision rule from the Economics chapters (Mankiw 10e: Ch 1, 2, 4, 5, 15), with what it is used for and how to apply it. Built for an open-note test: use the chapter headers to jump, and the Print button for a paper copy.

Ch 1 — Ten Principles: Decision Rules

RULE / EQUATION	WHAT IT IS USED FOR	HOW TO USE IT
Act if: Marginal Benefit > Marginal Cost (MB > MC)	Deciding whether to do <i>one more</i> of anything (watch one more movie, produce one more unit, stay open one more hour).	Compare only the extra benefit and extra cost of the increment. Ignore what is already spent or fixed. If MB > MC, do it; if MC > MB, don't.
Opportunity cost = value of the best alternative given up	Finding the TRUE cost of a choice (attending school, taking a job, using time or money one way instead of another).	Add up everything sacrificed — money spent <i>plus</i> foregone earnings/time. Example: the cost of college includes foregone wages, not just tuition.
Sunk cost: already spent → ignore it	Avoiding the sunk-cost trap in repair/replace/continue decisions.	Money already committed and unrecoverable (e.g., \$2,000 of past repairs) must NOT enter the decision. Compare only future MB vs. future MC.

Ch 2 — Thinking Like an Economist: Graphs & the PPF

EQUATION	WHAT IT IS USED FOR	HOW TO USE IT
Slope = $\Delta y \div \Delta x$ ("rise over run")	Measuring how steep a line is — how much y changes when x changes by one unit (e.g., along a demand curve).	Pick two points; divide the change in the y-value by the change in the x-value. Example from the text: moving from (13 novels, \$8) to (21, \$6): slope = $-2 \div 8 = -1/4$. Downward lines have negative slope; flat ≈ 0 ; vertical = infinite.
Opportunity cost on the PPF = amount of good B given up $\div$ amount of good A gained = slope of the PPF	Reading trade-offs off a production possibilities frontier.	Example: producing 100 more cars costs 200 computers → each car "costs" $200/100 = 2$ computers. A bowed-out PPF means this cost RISES as you make more of a good (specialized resources).
Points: on PPF = efficient · inside = inefficient · outside = not feasible	Classifying any output combination on a PPF question.	On the curve: can only get more of one good by giving up the other. Inside: could make more of BOTH. Outside: impossible with current resources/technology (growth shifts the whole curve outward).

Ch 4 — Supply & Demand: Market Math

EQUATION	WHAT IT IS USED FOR	HOW TO USE IT
Market demand = sum of all individual quantities demanded at each price	Building the market demand curve from individual buyers.	Add curves HORIZONTALLY (add quantities at each price, not prices). Example: at \$4, Catherine demands 4 + Nicholas demands 3 → market $Q_d = 7$. Market supply works the same way with sellers.
Equilibrium: Quantity demanded = Quantity supplied ($Q_d = Q_s$)	Finding the market-clearing price and quantity — where the curves intersect.	Read the intersection of supply and demand. At that price, buyers buy exactly what sellers sell; no pressure for price to move.
Surplus: $Q_s > Q_d$ (price above equilibrium) Shortage: $Q_d > Q_s$ (price below equilibrium)	Predicting price movement when the market is NOT in equilibrium.	Surplus → sellers cut prices → price falls to equilibrium. Shortage → sellers raise prices → price rises to equilibrium (the law of supply and demand).
3-step method: 1) Which curve shifts? 2) Which direction? 3) Compare old vs. new equilibrium	Analyzing ANY event's effect on price and quantity.	Results: $D \uparrow \rightarrow P \uparrow Q \uparrow$ · $D \downarrow \rightarrow P \downarrow Q \downarrow$ · $S \uparrow \rightarrow P \downarrow Q \uparrow$ · $S \downarrow \rightarrow P \uparrow Q \downarrow$. If BOTH curves shift, one of P or Q is ambiguous (depends on shift sizes).

Ch 5 — Elasticity: Formulas

EQUATION	WHAT IT IS USED FOR	HOW TO USE IT
Price elasticity of demand $= \% \Delta Q_d \div \% \Delta P$ (drop the minus sign)	Measuring how strongly buyers respond to a price change.	>1 elastic · <1 inelastic · $=1$ unit elastic · 0 perfectly inelastic (vertical curve) · ∞ perfectly elastic (horizontal curve). More elastic when: close substitutes exist, good is a luxury, market is narrowly defined, time horizon is long.
Midpoint method: $\% \Delta = (\text{end} - \text{start}) \div [(\text{end} + \text{start}) \div 2] \times 100$	Computing the percentage changes for ANY elasticity — gives the same answer in both directions.	Worked example (iPhones): P \$400→\$600, Q 10,600→8,400. $\% \Delta P = 200 \div 500 = 40\%$. $\% \Delta Q = 2,200 \div 9,500 \approx 23.2\%$. Elasticity = $23.2 \div 40 \approx 0.58 \rightarrow$ inelastic .
Total revenue = Price × Quantity (TR = P × Q)	What sellers receive / buyers spend; predicting revenue when price changes.	Inelastic demand: P and TR move TOGETHER (raise price → revenue rises). Elastic : P and TR move OPPOSITE (raise price → revenue falls). Unit elastic : TR unchanged. On a linear demand curve, elasticity falls as you slide down it: elastic at high P/low Q, inelastic at low P/high Q.
Income elasticity of demand $= \% \Delta Q_d \div \% \Delta \text{ income}$	Classifying goods by how demand responds to income.	Positive → normal good (well above 1 suggests a luxury). Negative → inferior good.
Cross-price elasticity $= \% \Delta Q_d \text{ of good 1} \div \% \Delta P \text{ of good 2}$	Determining whether two goods are substitutes or complements.	Positive → substitutes (pricier Coke → more Pepsi bought). Negative → complements (pricier peanut butter → less jelly bought).
Price elasticity of supply $= \% \Delta Q_s \div \% \Delta P$	Measuring how strongly sellers respond to a price change.	Same $>1/<1/=1$ classifications as demand. More elastic in the LONG run (capacity can adjust); very inelastic near maximum capacity. Use the midpoint method for the % changes.

Elasticity applications to remember: Farm technology → supply right + inelastic food demand → price falls more than quantity rises → farm revenue FALLS. OPEC → supply cuts spike prices short-run (inelastic) but barely move them long-run (elastic). Drug interdiction → supply left + inelastic demand → total drug spending RISES short-run.

Ch 15 — Competitive Firms: Cost & Revenue Formulas

EQUATION	WHAT IT IS USED FOR	HOW TO USE IT
$\text{Profit} = \text{TR} - \text{TC}$	The firm's bottom line and its goal (maximize this).	TR = total revenue, TC = total cost. Economic profit uses ALL opportunity costs (including the owner's time/capital) — so zero economic profit still means a positive accounting profit.
$\text{TR} = P \times Q \quad \cdot \quad \text{TC} = \text{FC} + \text{VC}$	Building blocks for every profit calculation.	FC = fixed costs (paid even at zero output); VC = variable costs (rise with output).
$\text{AR} = \text{TR} \div Q = P$ $\text{MR} = \Delta \text{TR} \div \Delta Q$	Average and marginal revenue.	AR equals price for ANY firm. For a COMPETITIVE (price-taking) firm, MR = P too — each extra unit adds exactly the market price to revenue.
$\text{ATC} = \text{TC} \div Q \quad \cdot \quad \text{AVC} = \text{VC} \div Q \quad \cdot \quad \text{AFC} = \text{FC} \div Q$ $\text{MC} = \Delta \text{TC} \div \Delta Q$	Per-unit cost measures used in every graph question.	Geometry to remember: MC slopes upward, ATC is U-shaped, and MC crosses ATC at ATC's MINIMUM.
Profit-max rule: produce where MR = MC (competitive firm: P = MC)	Choosing the profit-maximizing output.	If MR > MC → produce more; if MC > MR → produce less. Because the MC curve gives quantity supplied at each price, the MC curve IS the firm's supply curve.
Shutdown (short run): if P < AVC → produce 0 Exit (long run): if P < ATC Enter: if P > ATC	The stay-or-go decisions.	Short run: fixed costs are sunk, so only compare price to AVC. Three zones: P < AVC shut down now · AVC < P < ATC operate now but exit long run · P > ATC stay and profit. Short-run supply = MC above AVC; long-run supply = MC above ATC.
Profit = (P - ATC) × Q Loss = (ATC - P) × Q	Computing profit/loss and finding the rectangle on the graph.	At the chosen Q (where P = MC), the rectangle's height is P - ATC and width is Q. Example: Q = 30, P = \$3, ATC = \$5 → loss = (5 - 3) × 30 = \$60.
Long-run competitive equilibrium: P = minimum ATC → zero economic profit	Where free entry and exit push every competitive market.	P > ATC → profits → entry → supply right → price falls. P < ATC → losses → exit → price rises. Ends at P = min ATC with firms at efficient scale; long-run market supply is roughly horizontal there (and always more elastic than short-run).